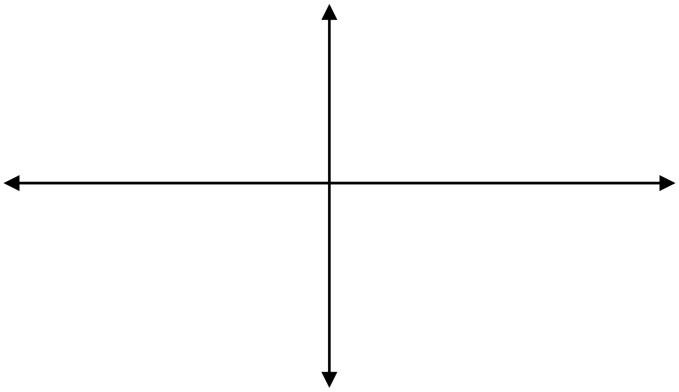
<u>**Chapter 10 Practice Quiz NC**</u>

1. Sketch the curve and indicate orientation. $x = t^3 - 4t \qquad y = 2t \qquad -2 \le t \le 2$



2. Find $\frac{dy}{dx}$ for $x = \sin\theta \quad y = \tan\theta$ at $\theta = \frac{\pi}{3}$.

3. Find the equation of the tangent line to the curve given by $x = 2t$ and $y = t^2 + 5$ at the point where $x = 2$.

4. <u>Set up</u> an integral that represents the arc length of the curve. $x = \cos(2t) \qquad y = \sin(2t) \qquad 0 \le t \le \frac{\pi}{2}$

5. A particle moves on a plane curve so that at any time t such that $t > 0$, its x - coordinate is $t^3 - t$ and its y - coordinate is $(2t-1)^3$. Find the acceleration vector of the particle at $t = 1$.

6. If $x = t^2$ and $y = \ln(t^2 + 1)$, then at $t = 1$, the value of $\frac{d^2y}{dx^2} =$

7. For what values of t does the curve given by the parametric equations $x = t^3 - t^2 - 1$ and $y = t^4 - 2t^2 - 8t$ have a vertical tangent?

8. Consider the curve in the xy-plane where the position is modeled by $\left(x(t), y(t)\right)$ and $\frac{dy}{dt} = \frac{10}{t^2 - 10t + 24}$. Given $y(1) = -2$, find the $y-$coordinate of the position at $t = 3$.

Free Response CA

A particle moving along a curve in the plane has position $(x(t), y(t))$ at time t, where

$$\frac{dx}{dt} = \sqrt{t^4 + 9} \text{ and } \frac{dy}{dt} = 2e^t + 5e^{-t}$$

for all real values of t. At time $t = 0$, the particle is at the point $(4, 1)$.

(a) Find the speed of the particle and its acceleration vector at time $t = 0$.

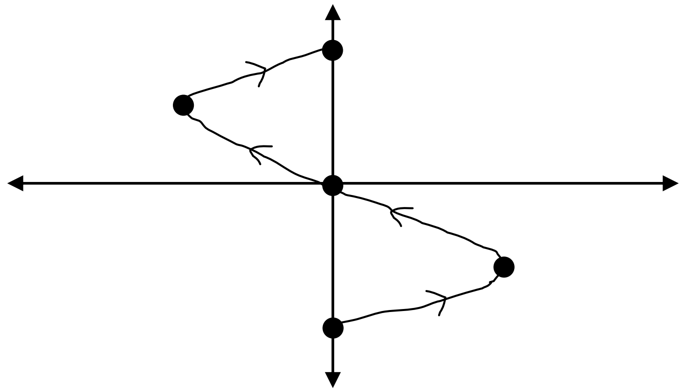
(b) Find an equation of the line tangent to the path of the particle at time $t = 0$.

(c) Find the total distance traveled by the particle over the time interval $0 \leq t \leq 3$.

(d) Find the x-coordinate of the position of the particle at time $t = 3$.

Answers

1. Create a table of values to plot some points:



t	x	y
-2	0	-4
-1	3	-2
0	0	0
1	-3	2
2	0	4

2. $\dfrac{dy}{dx}=\dfrac{dy/d\theta}{dx/d\theta}$, so $dy/d\theta=\sec^2\theta$ and $dx/d\theta=\cos\theta$. So, $\dfrac{dy}{dx}=\left.\dfrac{\sec^2\theta}{\cos\theta}\right|_{\theta=\frac{\pi}{3}}=\dfrac{4}{0.5}=8$.

3. We need point and slope to write the equation. Since $x=2$, we need t so $2=2t$, $t=1$ and so $y(1)=6$. The slope is $\dfrac{dy}{dx}=\dfrac{dy/dt}{dx/dt}=\dfrac{2t}{2}=t$, so the slope is 1. The equation would then be $y-6=1(x-2)$.

4. Arc length is $\displaystyle\int_0^{\frac{\pi}{2}}\sqrt{\left(\frac{dy}{dt}\right)^2+\left(\frac{dx}{dt}\right)^2}\,dt$, so $\dfrac{dy}{dt}=2\cos 2t$ and $\dfrac{dx}{dt}=-2\sin 2t$. So, $\displaystyle\int_0^{\frac{\pi}{2}}\sqrt{(2\cos 2t)^2+(-2\sin 2t)^2}\,dt$.

5. Given the position, we must differentiate twice to get acceleration. The position vector would be $\left\langle t^3-t,(2t-1)^3\right\rangle$ and the velocity vector would be $\left\langle 3t^2-1,6(2t-1)^2\right\rangle$ and the acceleration vector would be $\left\langle 6t,24(2t-1)\right\rangle$. At $t=1$, the acceleration vector would be $\langle 6,24\rangle$.

6. $\dfrac{d^2y}{dx^2}=\dfrac{\frac{d}{dt}\left[\frac{dy}{dx}\right]}{\frac{dx}{dt}}$ and $\dfrac{dy}{dx}=\dfrac{dy/dt}{dx/dt}$. Therefore, $\dfrac{dy}{dt}=\dfrac{2t}{t^2+1}$ and $\dfrac{dx}{dt}=2t$, so $\dfrac{dy}{dx}=\dfrac{\frac{2t}{t^2+1}}{2t}=\dfrac{1}{t^2+1}$. The second derivative would then be $\dfrac{d^2y}{dx^2}=\dfrac{\frac{d}{dt}\left[\frac{1}{t^2+1}\right]}{\frac{dx}{dt}}=\dfrac{\frac{-2t}{(t^2+1)^2}}{2t}=\left.\dfrac{-1}{(t^2+1)^2}\right|_{t=1}=-\dfrac{1}{4}$.

7. A vertical tangent line is when $\dfrac{dx}{dt}=0$. So, $\dfrac{dx}{dt}=3t^2-2t=0$ and $t(3t-2)=0$ or $t=0,\dfrac{2}{3}$.

8. Since we need to find the position and are given the velocity (derivative), then we need to integrate and add on the initial condition. So, $y(3) = -2 + \int_1^3 \frac{10}{t^2 - 10t + 24}\,dt$. Since this denominator is factorable, we should try partial fractions. $\frac{10}{t^2 - 10t + 24} = \frac{A}{t-4} + \frac{B}{t-6}$, $10 = A(t-6) + B(t-4)$ and we find that $A = -5$, $B = 5$, so $y(3) = -2 + \int_1^3 \frac{-5}{t-4}\,dt + \int_1^3 \frac{5}{t-6}\,dt$. This gives us $y(3) = -2 + \left[5\ln\left|\frac{t-6}{t-4}\right|\right]_1^3 = -2 + 5\ln 3 - 5\ln\frac{5}{3} = -2 + 5\ln\frac{9}{5}$

<u>Free Response CA</u>

(a) At time $t = 0$:

Speed $= \sqrt{x'(0)^2 + y'(0)^2} = \sqrt{3^2 + 7^2} = \sqrt{58}$

Acceleration vector $= \langle x''(0), y''(0)\rangle = \langle 0, -3\rangle$

$2 : \begin{cases} 1 : \text{speed} \\ 1 : \text{acceleration vector} \end{cases}$

(b) $\frac{dy}{dx} = \frac{y'(0)}{x'(0)} = \frac{7}{3}$

Tangent line is $y = \frac{7}{3}(x - 4) + 1$

$2 : \begin{cases} 1 : \text{slope} \\ 1 : \text{tangent line} \end{cases}$

(c) Distance $= \int_0^3 \sqrt{\left(\sqrt{t^4 + 9}\right)^2 + \left(2e^t + 5e^{-t}\right)^2}\,dt$
$= 45.226$ or 45.227

$3 : \begin{cases} 2 : \text{distance integral} \\ \quad \langle -1\rangle \text{ each integrand error} \\ \quad \langle -1\rangle \text{ error in limits} \\ 1 : \text{answer} \end{cases}$

(d) $x(3) = 4 + \int_0^3 \sqrt{t^4 + 9}\,dt$
$= 17.930$ or 17.931

$2 : \begin{cases} 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$